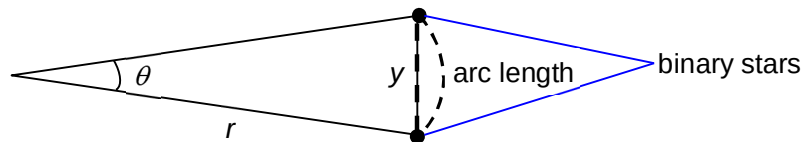


18

D



For just resolved image, $\sin \theta = \frac{\lambda}{b}$, where θ is very small,

$$\begin{aligned}\theta &\approx \frac{\lambda}{b} = \frac{620 \times 10^{-9}}{50 \times 10^{-2}} \\ &= 1.24 \times 10^{-6} \text{ rad}\end{aligned}$$

Distance between the two stars,

$$\begin{aligned}y &\approx r\theta \quad (\text{arc length}) \\ &= 5.7 \times 10^{16} \times 1.24 \times 10^{-6} \\ &= 7.068 \times 10^{10} \\ &\approx 7.1 \times 10^{10} \text{ m}\end{aligned}$$